Post-Stimulus Choice Decoding Is Stronger From Regional Firing Rates Than From Inter-Area Correlations

Shir Sivroni

Department of Mathematics and Computer Science, The Open University of Israel, Raanana 4310706, Israel

Correspondence: shirsi@openu.ac.il

Abstract

Neural decoders are often used to compare candidate representations of perceptual choice, but their interpretation is difficult when stimulus category, behavioral choice, and motor response are coupled by the task. I reanalyzed 39 simultaneous multi-region Neuropixels sessions from 10 mice performing a visual contrast-discrimination task. I compared two matched-dimensional feature families: regional mean firing rates and trial-wise inter-area correlations. Preprocessing was restricted to the training folds, performance was quantified by balanced accuracy, and inference was conducted on mouse-level summaries. Neither feature family decoded choice before stimulus onset (regional rates: 49.5%; inter-area correlations: 50.1%). After stimulus onset, regional firing-rate features were substantially more predictive than correlation features (0-250 ms: 64.2% versus 53.4%; 250-500 ms: 67.3% versus 50.1%; Holm-adjusted paired p = 0.0078 in both windows). The feature-family-by-time interaction was also reliable. Additional analyses showed that post-stimulus regional-rate decoding increased with stimulus discriminability and was much stronger on correct than error trials, indicating a substantial stimulus-linked component. Nevertheless, regional rates decoded choice above chance on equal-contrast trials, so stimulus side alone cannot explain the result. A decoder trained to identify stimulus side on correct trials did not generalize reliably to error trials after multiplicity correction. Thus, the data support a time-dependent difference in decodability from these two feature representations, not a causal switch from distributed to local computation. The results also illustrate why raw accuracy, fixed stimulus-response mappings, and model-dependent decoder performance must be handled explicitly when inferring neural mechanisms.

Significance Statement

After stimulus onset, a brain-wide vector of regional firing rates predicts the mouse's left/right response more accurately than an equally sized set of trial-wise correlations between regions. This advantage is absent before the stimulus and remains when trials are evaluated before movement onset. However, the advantage is largest for easy and correct trials, where stimulus and response are most tightly coupled. The findings therefore establish a difference between feature representations while placing clear limits on claims about localization, communication, or causal decision mechanisms.

Keywords: perceptual decision-making; Neuropixels; neural decoding; inter-area correlation; firing rate; cross-validation; stimulus-choice confounding

Introduction

Perceptual decisions are associated with activity distributed across sensory, frontal, and motor systems (Shadlen and Newsome, 2001; Gold and Shadlen, 2007; Hanks and Summerfield, 2017; Steinmetz et al., 2019). Ongoing activity before stimulus onset can covary with subsequent perception or response (Arieli et al., 1996; Hesselmann et al., 2008; Wyart and Tallon-Baudry, 2009; Meijer et al., 2023), whereas post-stimulus population activity reflects sensory input, movement, engagement, and choice (Musall et al., 2019; Steinmetz et al., 2019; Stringer et al., 2019; Zatka-Haas et al., 2021; Musall and Churchland, 2024). An important methodological question is whether these periods are better characterized by relations between regions or by firing-rate signals summarized within regions.

Relations between regions can be represented in many ways, including zero-lag covariation, frequency-specific coherence, lagged dependence, and directed or nonlinear interactions (Fries, 2005, 2015; Siegel et al., 2012). Choice-related population activity can likewise reflect recurrent dynamics and mixed sensory, contextual, and motor variables (Mante et al., 2013; Panzeri et al., 2017). Pre-stimulus variation may further covary with arousal and choice history (McGinley et al., 2015; Urai et al., 2019) or with emerging motor-cortical choice signals (Donner et al., 2009). The present comparison therefore concerns two explicitly defined feature constructions, not an exhaustive contrast between “network” and “local” neural computation.

Decoder performance can compare feature representations, but it does not directly identify the biological mechanism that generated them. Accuracy is a lower bound on the information accessible to a particular model and depends on feature construction, dimensionality, regularization, preprocessing, class balance, and cross-validation; model weights and prediction performance also require cautious biological interpretation (Haufe et al., 2014; Varoquaux et al., 2017). Moreover, a feature vector of regional firing rates is still brain-wide when it concatenates rates from multiple regions, and a zero-lag correlation calculated within a short window need not measure communication or causal connectivity. These distinctions limit inferences from a decoder's success or failure.

The behavioral design adds a second interpretive problem. In the Steinmetz et al. task, mice turned a wheel to bring the higher-contrast stimulus to the center. On unequal-contrast correct trials, stimulus side determines the rewarded wheel direction; stimulus category, correct choice, and motor response are therefore deterministically linked. High post-stimulus choice decoding may consequently reflect sensory coding, response preparation, movement, or a mixture. Error trials, equal-contrast trials, stimulus difficulty, and movement timing are necessary controls.

Here I reanalyzed the public dataset using a leakage-resistant, class-balanced, mouse-level framework. The primary analysis compares matched-dimensional regional firing-rate and inter-area correlation features across four time windows. The secondary analyses decode stimulus side, stratify choice decoding by difficulty and correctness, evaluate equal-contrast trials, test correct-to-error generalization, and restrict evaluation to trials whose movement began after the analyzed window. The aim is to identify what the recorded data support while avoiding mechanistic conclusions that the analysis cannot establish.

Materials and Methods

Dataset and behavioral task

I analyzed the publicly available Steinmetz et al. (2019) dataset: 39 Neuropixels recording sessions from 10 mice. Visual stimuli of independently varied contrast appeared on the left and right. The mouse reported its decision by turning a wheel so that the higher-contrast stimulus moved toward the center. The primary analysis included 6,745 trials with a leftward or rightward response (median, 161 per session; range, 63-329). No-response trials were excluded. For unequal contrasts, signed stimulus side was defined by the sign of right minus left contrast. Correct and error trials were defined by the dataset's feedback variable.

Time windows and regional population signals

Spikes were represented in 10-ms bins with stimulus onset at bin 50. Four non-overlapping 250-ms windows were analyzed: -500 to -250 ms (pre-early), -250 to 0 ms (pre-late), 0 to 250 ms (post-early), and 250 to 500 ms (post-late). Regions containing at least 10 recorded neurons were retained.

For each retained region and trial, the regional firing-rate feature was the mean spike count across neurons and time bins in the window. If a session contained R retained regions, the regional-rate decoder used an R-dimensional vector.

For each pair of retained regions, I first averaged activity across neurons to obtain two regional population time series. Their Pearson correlation across the 25 time bins yielded one trial-wise inter-area feature. To match dimensionality, each correlation decoder used R region-pair features. Because more than R pairs were usually available, results were averaged over 10 reproducible random subsets of R pairs. This avoids comparing feature matrices of different size or relying on an arbitrary first-ten-pairs selection.

Decoding and cross-validation

Choice and stimulus side were decoded with L2-regularized logistic regression. Models used inverse-frequency class weighting. Performance was quantified by balanced accuracy, the mean of the two class-specific recall values, for which chance performance is 0.5 even when the numbers of left and right choices differ.

Each session used stratified five-fold cross-validation with shuffled, reproducible folds. Standardization was implemented inside a pipeline and fitted separately on each training fold. Thus, test-fold observations did not contribute to feature means or variances. Out-of-fold predictions from the primary choice decoder were also evaluated within prespecified trial strata.

Controls requested during review

Stimulus decoding. On unequal-contrast trials, the same feature families were used to decode whether the higher contrast appeared on the left or right.

Difficulty. Trials were grouped by absolute contrast difference: hard (0.25), medium (0.50), and easy (at least 0.75). Choice accuracy was evaluated within each group using out-of-fold predictions from the all-trial model.

Correct and error trials. Out-of-fold choice predictions were evaluated separately on correct and error trials. In a complementary analysis, a stimulus-side decoder was trained only on correct unequal-contrast trials and tested on error unequal-contrast trials. Because the stimulus-response relation reverses on errors, above-chance generalization would indicate stimulus-following activity, whereas below-chance performance would indicate response-following activity.

Equal-contrast trials. Choice was decoded using only trials in which left and right contrasts were equal. This removes signed stimulus side as an explanation, although it does not remove motor or other choice-correlated signals.

Pre-movement evaluation. For the post-early and post-late windows, performance was evaluated separately on trials whose recorded reaction time was at least 250 or 500 ms, respectively. In those subsets, movement began after the analyzed window.

Statistical inference

Session-level balanced accuracies were averaged within mouse, making the mouse (n = 10) the inferential unit. Means and 95% confidence intervals were obtained by resampling mice with replacement (20,000 bootstrap samples). Differences were tested with exact two-sided sign-flip tests over mouse-level paired effects. Holm correction was applied within each prespecified family of comparisons. The smallest attainable two-sided exact p value with 10 mice is 0.001953.

Code and data availability

The source dataset is available with Steinmetz et al. (2019). The original analysis repository is available at https://github.com/shir-openu/steinmetz-neural-decision-signals. The revised analysis script accompanies this manuscript and uses a fixed random seed (20260720).

Ethics statement

This study is a secondary analysis of publicly available data and involved no new animal experiments. The experimental procedures and ethical approvals governing the original recordings are reported in Steinmetz et al. (2019).

Results

Balanced reanalysis does not support pre-stimulus choice decoding

The original analysis used ordinary accuracy and tested it against 50%. However, choices were imbalanced within sessions: the mean mouse-level majority-class baseline was 58.5% (95% CI, 56.7-60.2%). Consequently, raw accuracies near 54% cannot establish above-chance prediction. With balanced accuracy, leakage-resistant preprocessing, matched feature dimensions, and mouse-level inference, neither feature family decoded choice before stimulus onset. In the pre-late window, regional-rate accuracy was 49.5% (95% CI, 47.7-51.3%) and inter-area correlation accuracy was 50.1% (49.2-50.9%); neither differed from chance, and their paired difference was not significant (Table 1; Fig. 1A).

Regional firing-rate features become strongly predictive after stimulus onset

After stimulus onset, regional-rate features outperformed matched-dimensional inter-area correlation features. In the 0-250 ms window, balanced accuracy was 64.2% for regional rates and 53.4% for correlations, a paired difference of 10.9 percentage points (95% CI, 8.5-13.2; Holm-adjusted p = 0.0078). In the 250-500 ms window, the corresponding values were 67.3% and 50.1%, a difference of 17.2 points (14.4-20.3; adjusted p = 0.0078).

The critical interaction compared the local-minus-correlation difference after stimulus onset with the same difference in the pre-late window. It was 11.5 points for post-early (8.3-14.9) and 17.8 points for post-late (13.8-22.2); exact p = 0.001953 for each, Holm-adjusted p = 0.0117 across the temporal-contrast family. The supported temporal result is therefore the emergence of a relative advantage for regional-rate features, not pre-stimulus decoding by inter-area correlations.

Table 1. Mouse-level balanced choice-decoding accuracy across time windows

Window	Regional firing rates	Inter-area correlations	Difference	Holm-adjusted paired p
Pre-early (-500 to -250 ms)	49.2% [47.1-51.3]	50.3% [49.4-51.3]	-1.1% [-3.0, 0.7]	0.609
Pre-late (-250 to 0 ms)	49.5% [47.7-51.3]	50.1% [49.2-50.9]	-0.6% [-2.5, 1.3]	0.609
Post-early (0 to 250 ms)	64.2% [61.5-67.1]	53.4% [52.2-54.4]	10.9% [8.5-13.2]	0.0078
Post-late (250 to 500 ms)	67.3% [64.2-70.5]	50.1% [48.9-51.3]	17.2% [14.4-20.3]	0.0078

Values are means across 10 mouse-level summaries with bootstrap 95% confidence intervals. Difference is regional firing rates minus inter-area correlations. Tests are exact paired sign-flip tests with Holm correction across the four windows.

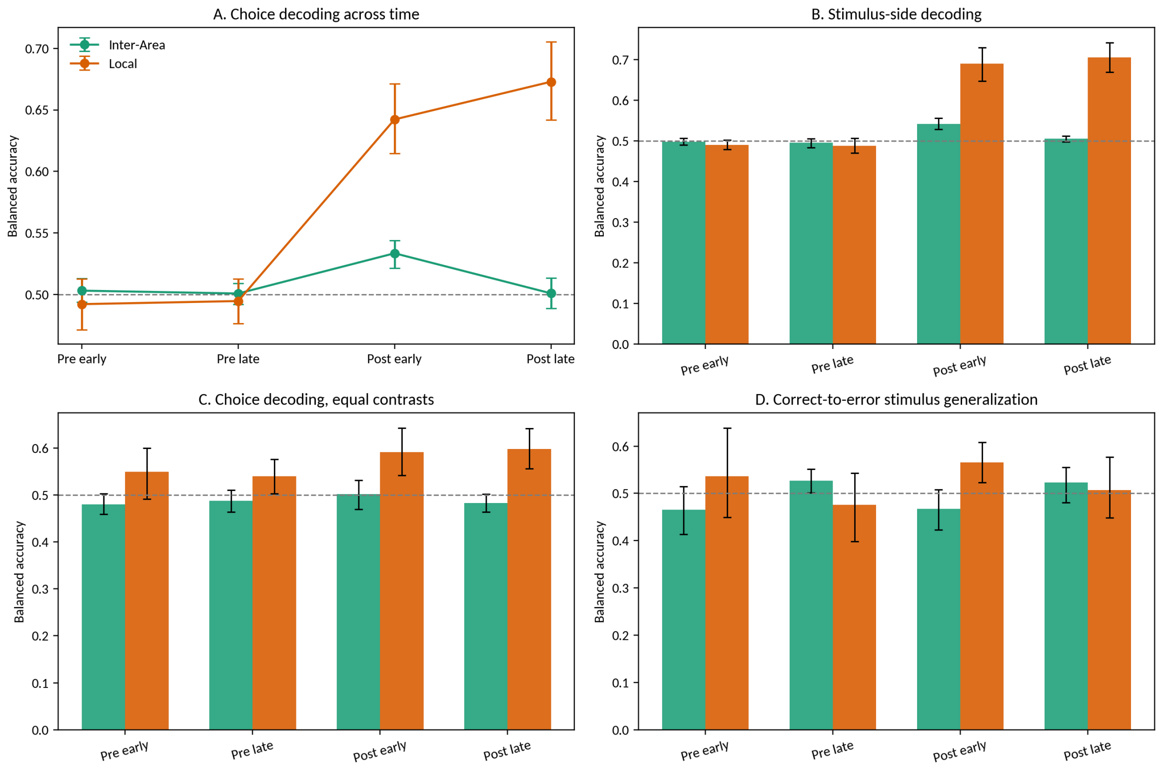

Figure 1. Time-dependent choice and stimulus decoding. (A) Choice decoding from matched-dimensional inter-area correlation and regional firing-rate features. Error bars are mouse-level bootstrap 95% confidence intervals. (B) Stimulus-side decoding on unequal-contrast trials. (C) Choice decoding restricted to equal-contrast trials. (D) Stimulus decoder trained on correct trials and tested on error trials. The dashed line marks chance balanced accuracy (0.5).

Stimulus discriminability and behavioral correctness account for substantial post-stimulus decoding

Regional firing rates decoded stimulus side at 69.0% in the post-early window and 70.5% in the post-late window (Fig. 1B). Choice decoding from regional rates also increased with stimulus discriminability. In the post-early window it rose from 59.5% on hard trials to 67.0% on medium trials and 70.5% on easy trials. The easy-minus-hard difference was 11.0 points (95% CI, 6.3-15.6; family-adjusted p = 0.0117). In the post-late window the corresponding difference was 5.9 points (1.3-10.3; adjusted p = 0.0449).

The effect of correctness was larger. Post-early regional-rate accuracy was 67.7% on correct trials but 53.9% on error trials, a difference of 13.8 points (8.2-19.2; adjusted p = 0.0117). Post-late accuracy was 71.3% on correct trials and 54.4% on errors, a difference of 16.9 points (11.3-22.5; adjusted p = 0.0078). Neither error-trial regional-rate accuracy remained significantly above chance after Holm correction within the correctness family. These patterns show that the strong all-trial post-stimulus result contains a major component tied to the task's stimulus-response mapping.

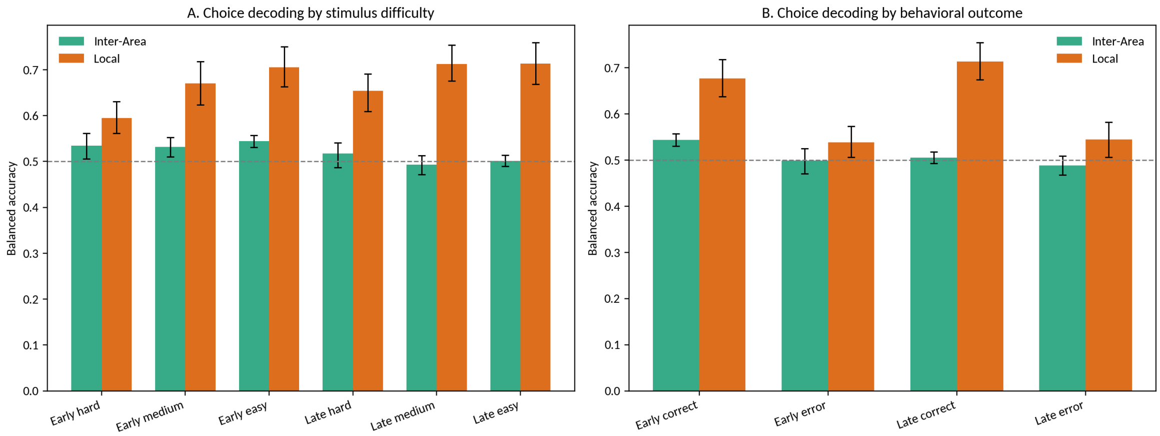

Figure 2. Difficulty and error-trial controls. (A) Choice-decoding balanced accuracy by absolute contrast difference. (B) Choice-decoding balanced accuracy evaluated separately on correct and error trials. Models were trained with all eligible left/right response trials, and plotted values evaluate their out-of-fold predictions within each stratum. Error bars are mouse-level bootstrap 95% confidence intervals.

Stimulus side does not fully explain the regional-rate signal

When left and right contrasts were equal, signed stimulus side was absent. Regional firing rates nevertheless decoded choice at 59.1% in the post-early window (95% CI, 54.2-64.2%; Holm-adjusted p = 0.0234) and 59.8% in the post-late window (55.6-64.2%; adjusted p = 0.0078). Inter-area correlation performance remained at chance. Thus, the regional-rate advantage cannot be reduced to stimulus side alone; it may also contain choice, motor preparation, movement, internal-state, or other response-correlated signals.

The correct-to-error generalization test was inconclusive. A regional-rate stimulus decoder trained on correct trials reached 56.6% on post-early error trials, but this did not survive correction across time windows (adjusted p = 0.125); post-late performance was 50.7%. Error trials were much less numerous than correct trials, limiting power. The present data therefore do not cleanly identify whether the residual post-stimulus signal follows stimulus identity or the animal's response on error trials.

The regional-rate advantage is present before movement onset

Restricting evaluation to trials whose recorded movement began only after the analyzed window retained the post-stimulus regional-rate advantage (Table 2). In the post-early window, regional-rate accuracy was 60.5% and correlation accuracy was 51.8%, a difference of 8.6 points (95% CI, 5.5-11.5; Holm-adjusted p = 0.0078). In the post-late window, the corresponding values were 54.4% and 49.2%, a difference of 5.2 points (2.1-8.1; adjusted p = 0.0137). This control reduces, but does not eliminate, the possibility that preparatory motor activity or other movement-correlated signals contribute before overt movement.

Table 2. Pre-movement choice decoding

Window and restriction	Regional firing rates	Inter-area correlations	Difference	Holm-adjusted paired *p*
Post-early; reaction time at least 250 ms	60.5% [57.4-63.6]	51.8% [49.9-53.3]	8.6% [5.5-11.5]	0.0078
Post-late; reaction time at least 500 ms	54.4% [51.3-57.1]	49.2% [47.7-50.5]	5.2% [2.1-8.1]	0.0137

Models were trained and cross-validated on all eligible left/right response trials; balanced accuracy was then evaluated on out-of-fold predictions from trials whose reaction time exceeded the end of the relevant window.

Discussion

The reanalysis supports a clear but narrower conclusion than the original manuscript. Before stimulus onset, neither regional firing rates nor trial-wise inter-area correlations reliably decode choice. After stimulus onset, regional firing-rate features become substantially more predictive than matched-dimensional correlation features. The primary inferential result is a feature-family-by-time interaction: the relative advantage of regional rates emerges after sensory information becomes available.

This result does not demonstrate a switch from a distributed neural mechanism to a local one. The regional-rate model concatenates signals from multiple regions and is therefore itself brain-wide. Conversely, a trial-wise zero-lag correlation is only one summary of inter-area relations and may be noisy when estimated from 25 time bins. Failure of this correlation representation to decode choice is not evidence that inter-area communication is absent. Decoder accuracy is model- and representation-dependent, and it provides a lower bound rather than a complete inventory of encoded information.

The task-mapping controls materially constrain the interpretation of post-stimulus decoding. Regional-rate choice decoding is greater for easy than hard trials and for correct than error trials, while regional rates directly decode stimulus side. These findings are expected when neural responses carry sensory information and the correct response is fixed by stimulus category. They preclude describing all post-stimulus accuracy as choice-specific evidence accumulation. At the same time, above-chance decoding on equal-contrast trials shows that signed stimulus side is not the whole explanation. The residual could reflect the animal's choice, planned or executed movement, engagement, or correlated internal state. Zatka-Haas et al. (2021) similarly distinguished localized sensory correlates from widespread movement correlates and emphasized that correlation alone does not establish causal relevance.

The revised analysis also identifies why the earlier pre-stimulus claim appeared stronger. Ordinary accuracy was tested against 50% even though within-session choices were imbalanced. The majority-class baseline averaged 58.5% across mouse-level summaries, higher than the reported pre-stimulus accuracies. Balanced accuracy and mouse-level inference remove this artifact. Fitting feature standardization within training folds and comparing equal-dimensional feature sets further reduce optimistic or inequitable comparisons.

Limitations

First, the analysis remains observational and cannot establish causal contribution. Second, region-level averaging discards neuron-specific population geometry, and the chosen correlation measure discards lagged, spectral, directional, and nonlinear interactions. It is therefore much narrower than communication-through-coherence or spectral-interaction frameworks (Fries, 2005, 2015; Siegel et al., 2012). Third, correlation values estimated from 250-ms windows can be unstable, so the comparison concerns these concrete feature constructions rather than all local and distributed codes. Fourth, the correct-to-error analysis is underpowered because error trials are relatively rare. Fifth, the pre-movement restriction uses recorded reaction time and cannot exclude covert motor preparation. Finally, feature subsets and decoder hyperparameters were fixed rather than optimized; this protects against circularity but may underestimate either representation's attainable performance.

Conclusions

Regional firing-rate features, aggregated across recorded regions, show a robust post-stimulus advantage over matched-dimensional trial-wise inter-area correlations for decoding the animal's response. The advantage is absent before the stimulus, is present before overt movement, and is strongly modulated by stimulus difficulty and behavioral correctness. The appropriate interpretation is a time-dependent difference in decodability from two feature representations. The data do not establish a dual-mechanism model, localization of the decision, or a causal transition from network coordination to local computation.

References

Arieli A, Sterkin A, Grinvald A, Aertsen A (1996) Dynamics of ongoing activity: explanation of the large variability in evoked cortical responses. *Science* 273:1868-1871.

Donner TH, Siegel M, Fries P, Engel AK (2009) Buildup of choice-predictive activity in human motor cortex during perceptual decision making. *Current Biology* 19:1581-1585.

Fries P (2005) A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends in Cognitive Sciences* 9:474-480.

Fries P (2015) Rhythms for cognition: communication through coherence. *Neuron* 88:220-235.

Gold JI, Shadlen MN (2007) The neural basis of decision making. *Annual Review of Neuroscience* 30:535-574.

Hanks TD, Summerfield C (2017) Perceptual decision making in rodents, monkeys, and humans. *Neuron* 93:15-31.

Haufe S, Meinecke F, Gorgen K, Dahne S, Haynes JD, Blankertz B, Biessmann F (2014) On the interpretation of weight vectors of linear models in multivariate neuroimaging. *NeuroImage* 87:96-110.

Hesselmann G, Kell CA, Eger E, Kleinschmidt A (2008) Spontaneous local variations in ongoing neural activity bias perceptual decisions. *Proceedings of the National Academy of Sciences USA* 105:10984-10989.

Mante V, Sussillo D, Shenoy KV, Newsome WT (2013) Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature* 503:78-84.

McGinley MJ, Vinck M, Reimer J, Batista-Brito R, Zagha E, Cadwell CR, Tolias AS, Cardin JA, McCormick DA (2015) Waking state: rapid variations modulate neural and behavioral responses. *Neuron* 87:1143-1161.

Meijer GT, Montijn JS, Bhattacharyya M, et al. (2023) Pre-stimulus brain states predict perceptual decisions. *Nature Neuroscience* 26:2042-2054.

Musall S, Kaufman MT, Juavinett AL, Gluf S, Churchland AK (2019) Single-trial neural dynamics are dominated by richly varied movements. *Nature Neuroscience* 22:1677-1686.

Musall S, Churchland AK (2024) Spontaneous and task-evoked dynamics in neural populations. *Annual Review of Neuroscience* 47:447-473.

Panzeri S, Harvey CD, Piasini E, Latham PE, Fellin T (2017) Cracking the neural code for sensory perception by combining statistics, intervention, and behavior. *Neuron* 93:491-507.

Shadlen MN, Newsome WT (2001) Neural basis of a perceptual decision in the parietal cortex (area LIP) of the rhesus monkey. *Journal of Neurophysiology* 86:1916-1936.

Siegel M, Donner TH, Engel AK (2012) Spectral fingerprints of large-scale neuronal interactions. *Nature Reviews Neuroscience* 13:121-134.

Steinmetz NA, Zatka-Haas P, Carandini M, Harris KD (2019) Distributed coding of choice, action and engagement across the mouse brain. *Nature* 576:266-273.

Stringer C, Pachitariu M, Steinmetz N, Reddy CB, Carandini M, Harris KD (2019) Spontaneous behaviors drive multidimensional, brainwide activity. *Science* 364:eaav7893.

Urai AE, Braun A, Donner TH (2019) Pupil-linked arousal is driven by decision uncertainty and alters serial choice bias. *Nature Communications* 10:4371.

Varoquaux G, Raamana PR, Engemann DA, Hoyos-Idrobo A, Schwartz Y, Thirion B (2017) Assessing and tuning brain decoders: cross-validation, caveats, and guidelines. *NeuroImage* 145:166-179.

Wyart V, Tallon-Baudry C (2009) How ongoing fluctuations in human visual cortex predict perceptual awareness: baseline shift versus decision bias. *Journal of Neuroscience* 29:8715-8725.

Zatka-Haas P, Steinmetz NA, Carandini M, Harris KD (2021) Sensory coding and the causal impact of mouse cortex in a visual decision. *eLife* 10:e63163.

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Declaration of Interests

The author declares no competing interests.

Supplementary Materials: Reproducible Analysis Specification

S1. Data Structure and Notation

S1.1 Neural Data Tensor

For each recording session, the preprocessed spike-count data are represented as a three-dimensional array

$$\mathbf{S} \in \mathbb{R}^{K \times N \times T}, \quad \text{(S1.1)}$$

where K is the number of trials, N is the number of recorded neurons, and T = 250 is the number of 10-ms bins spanning -500 to 2,000 ms relative to stimulus onset. The element $S_{k,n,t}$ is the spike count for trial k, neuron n, and time bin t. The equations below use the equivalent notation $S_{n,k,t}$ when neuron is written as the first conceptual index.

S1.2 Brain-Region Assignment

Each neuron is assigned to one anatomical label by a mapping $\alpha:\{1,...,N\} \rightarrow A$. For region a, the neuron index set and its size are

$$I_a = \{n : \alpha(n) = a\}, \quad N_a = |I_a|. \quad \text{(S1.2)}$$

A region is retained when $N_a \geq 10$. A session is eligible for the matched-dimensional comparison when it contains at least three retained regions.

S1.3 Time Windows

Stimulus onset occurs at bin t_0 = 50. The four non-overlapping windows are:

Window	Time range	Bin range	Role in analysis
Pre-early	-500 to -250 ms	[0,25)	Early baseline
Pre-late	-250 to 0 ms	[25,50)	Immediately before stimulus
Post-early	0 to 250 ms	[50,75)	Early stimulus/response period
Post-late	250 to 500 ms	[75,100)	Later stimulus/response period

S1.4 Behavioral Variables

The primary binary response variable is

$$y_k = 1 \text{ for a rightward response}; \quad y_k = 0 \text{ for a leftward response}. \quad \text{(S1.3)}$$

No-response trials are excluded. Signed stimulus side is $g_k = \text{sign}(c_{\text{right},k} - c_{\text{left},k})$, where c denotes contrast. Correctness is obtained from the feedback variable, and difficulty is $|c_{\text{right},k} - c_{\text{left},k}|$.

S2. Feature Computation

S2.1 Population Activity

Let $S_{n,k,t}$ denote the spike count of neuron n on trial k in 10-ms time bin t. Each neuron is assigned to a recorded brain region. For region a, let I_a be the index set of its neurons and let $N_a = |I_a|$. Only regions with $N_a \geq 10$ were retained. For a decoding window $W = [t_1, t_2)$, the regional population activity is the neuron-averaged time series

$$p_{a,k}(t) = (1 / N_a)\, \Sigma_{n \in I_a} S_{n,k,t}, \quad t \in W. \quad \text{(S2.1)}$$

Thus, for every region-trial pair, $\mathbf{p}_{a,k}$ is a vector of $|W| = 25$ observations. The averaging is performed separately within every trial and window; activity from other trials does not enter this quantity.

S2.2 Inter-Area Correlation

For two retained regions a and b, the inter-area feature on trial k is the Pearson correlation between their regional population time series across the 25 bins in the window:

$$\rho_{ab,k} = \Sigma_{t \in W} [p_{a,k}(t) - \bar{p}_{a,k}][p_{b,k}(t) - \bar{p}_{b,k}] / \{\sqrt{(\Sigma_{t \in W}[p_{a,k}(t) - \bar{p}_{a,k}]^2)}\ \sqrt{(\Sigma_{t \in W}[p_{b,k}(t) - \bar{p}_{b,k}]^2)}\}. \quad \text{(S2.2)}$$

Here,

$$\bar{p}_{a,k} = (1 / |W|)\, \Sigma_{t \in W}\, p_{a,k}(t). \quad \text{(S2.3)}$$

If either regional time series is constant within a window, the denominator in (S2.2) is zero and the correlation feature is set to zero, matching the original implementation.

If a session contains R retained regions, there are $P = R(R - 1)/2$ possible unordered region pairs. To compare the two feature families at the same dimensionality, the corrected analysis does not use all P correlations. For repetition s, it selects without replacement a fixed-seed subset Q_s of exactly R region pairs and constructs

$$\mathbf{x}^{\text{inter},(s)}{}_k = (\rho_{ab,k} : (a,b) \in Q_s)^T \in \mathbb{R}^R, \quad |Q_s| = R. \quad \text{(S2.4)}$$

Ten reproducible subsets (s = 1,...,10) are evaluated and their decoder scores are averaged. This prevents the comparison from being driven by unequal feature counts or by an arbitrary alphabetical selection of region pairs. At least three retained regions are required because $P \geq R$ only when $R \geq 3$.

S2.3 Regional Firing-Rate Activity

For region a, the regional firing-rate feature is the mean spike count across its neurons and across the time bins in the window:

$$r_{a,k} = [1 / (N_a|W|)]\, \Sigma_{n \in I_a} \Sigma_{t \in W} S_{n,k,t} = (1 / |W|)\, \Sigma_{t \in W}\, p_{a,k}(t). \quad \text{(S2.5)}$$

For a session with R retained regions, the regional firing-rate vector is

$$\mathbf{x}^{regional}_k = (r_{a1,k}, r_{a2,k}, ..., r_{aR,k})^T \in R^R. \quad (S2.6)$$

Equations (S2.4) and (S2.6) therefore yield exactly *R* predictors per trial for each feature family. The terms “regional firing-rate” and “local” refer to this same feature construction; because the vector concatenates multiple regions, it remains a brain-wide representation and should not be interpreted as a single-region or causal-local measure.

S3. Classification Framework

S3.1 Logistic Regression

For either feature vector $\mathbf{x}_k \in R^R$, the probability of a rightward response is modeled as

$$P(y_k = 1 \mid \mathbf{x}_k) = \sigma(\mathbf{w}^T\mathbf{x}_k + b) = 1 / \{1 + \exp[-(\mathbf{w}^T\mathbf{x}_k + b)]\}. \quad (S3.1)$$

The fitted classifier is L2-regularized logistic regression with an intercept, *C* = 1, the liblinear solver, a maximum of 2,000 iterations, and random seed 20260720.

S3.2 Weighted Regularized Objective

To reduce the effect of unequal left/right response counts, the loss assigns inverse-frequency weights to the two response classes. In schematic form, the fitted parameters minimize

$$L(\mathbf{w},b) = -\Sigma_{k \in \text{train}}\, \omega_{y_k}\{y_k\log(\hat{p}_k) + (1-y_k)\log(1-\hat{p}_k)\} + \lambda||\mathbf{w}||_2^2, \quad (S3.2)$$

where $\hat{p}_k$ is given by (S3.1), ω_c is inversely proportional to the number of training observations in class *c*, and the effective regularization strength is controlled by *C* = 1.

S3.3 Fold-Specific Standardization

For every cross-validation fold and feature *j*, the training-fold mean and standard deviation are

$$\mu_{j,f} = (1/|\text{Train}_f|)\Sigma_{k \in \text{Train}_f} x_{k,j}, \quad \sigma_{j,f}^2 = (1/|\text{Train}_f|)\Sigma_{k \in \text{Train}_f}(x_{k,j}-\mu_{j,f})^2. \quad (S3.3)$$

Both training and held-out observations are transformed using these training-fold quantities:

$$z_{k,j,f} = (x_{k,j} - \mu_{j,f}) / \sigma_{j,f}. \quad (S3.4)$$

The held-out fold therefore contributes neither to feature centering nor to scaling. Standardization and logistic regression are combined in one scikit-learn pipeline.

S3.4 Stratified Cross-Validation and Balanced Accuracy

Within each session, trials are divided into five shuffled, stratified folds. When a secondary subset contains too few observations, the number of folds is reduced to the smaller class count; at least three folds and 30 eligible trials are required. Each trial receives one prediction from a model that was not trained on that trial.

For true positive count TP, false negative count FN, true negative count TN, and false positive count FP, the fold-combined score is

$$BA = 0.5[TP/(TP+FN) + TN/(TN+FP)]. \quad (S3.5)$$

Balanced accuracy equals 0.5 for a chance classifier even when the response classes are unequal. For the inter-area representation, (S3.5) is computed for each of the 10 matched pair subsets and then averaged.

S4. Statistical Inference

S4.1 Mouse-Level Aggregation

Sessions from the same mouse are not treated as independent replicates. If mouse *m* contributes the set of sessions J_m, its summary score for feature family *f* and condition *c* is

$$A_{m,f,c} = (1/|J_m|)\Sigma_{j \in J_m} BA_{j,f,c}. \quad (S4.1)$$

The reported mean is the unweighted mean of the 10 mouse-level values. Paired feature-family effects are $d_{m,c} = A_{m,regional,c} - A_{m,inter,c}$. Temporal interactions compare this paired difference after stimulus onset with the corresponding pre-late difference.

S4.2 Bootstrap Confidence Intervals

Confidence intervals are percentile intervals from 20,000 bootstrap samples of mice. Each bootstrap sample draws 10 mice with replacement and recomputes the mouse-level mean or paired mean difference. The 2.5th and 97.5th percentiles form the 95% interval.

S4.3 Exact Paired Sign-Flip Test

For a paired mouse-level effect vector $\mathbf{d} = (d_1,...,d_M)$, the two-sided exact null distribution enumerates all 2^M sign assignments:

$$p = 2^{-M}\Sigma_{\mathbf{s} \in \{-1,+1\}^M} 1[|(1/M)\Sigma_{m=1}^M s_m d_m| \geq |\bar{d}|]. \quad (S4.2)$$

For a comparison with chance, $d_m = A_{m,f,c} - 0.5$. With $M = 10$ mice, the smallest attainable two-sided exact p value is $2/2^{10} = 0.001953$.

S4.4 Multiplicity Correction

Holm correction is applied within each prespecified analysis family. If $p_{(1)} \leq ... \leq p_{(q)}$ are the ordered raw values, then

$$p_{(i),adj} = \max_{j\leq i}\{\min[1,(q-j+1)p_{(j)}]\}. \quad (S4.3)$$

Separate families are used for the four primary windows, the six temporal contrasts, stimulus decoding, equal-contrast decoding, correct-to-error generalization, difficulty, correctness, and pre-movement evaluation.

S5. Analysis Algorithm

1. Load all 39 sessions and retain trials with a leftward or rightward response.
2. For each session, identify regions containing at least 10 neurons; require at least three retained regions.
3. For each of the four windows, compute the region-by-time population signals using (S2.1).
4. Compute all trial-wise pair correlations using (S2.2) and regional firing rates using (S2.5).
5. Create 10 fixed-seed subsets of R inter-area features; use the same R-dimensional regional-rate vector for the comparison.
6. Generate shuffled stratified folds and obtain leakage-resistant out-of-fold predictions from the scaler-plus-logistic-regression pipeline.
7. Compute balanced accuracy for choice, stimulus side, and every prespecified trial stratum.
8. Average session scores within each mouse.
9. Compute mouse-level means, bootstrap confidence intervals, exact paired sign-flip tests, and Holm-adjusted p values.
10. Write the complete session-level and summary results to JSON and generate the manuscript figures.

S6. Implementation and Reproducibility

Component	Implementation	Specification
Correlation	NumPy Pearson correlation	25 time bins per trial and region pair
Classifier	scikit-learn LogisticRegression	L2, C=1, class_weight=balanced, liblinear
Pipeline	StandardScaler + classifier	Fitted within training folds
Cross-validation	StratifiedKFold	Five shuffled folds; seed 20260720
Primary score	balanced_accuracy_score	Chance = 0.5
Confidence interval	NumPy bootstrap	20,000 mouse resamples
Hypothesis test	Exact enumeration	Two-sided paired sign flips
Multiplicity	Holm procedure	Within prespecified families

Environment: Python 3.12.13; NumPy 2.3.5; SciPy 1.17.0; scikit-learn 1.8.0; Matplotlib 3.10.8. The complete executable analysis is provided in analysis_additional.py, and the complete numerical output is provided in reanalysis_results.json.

Code availability: https://github.com/shir-openu/steinmetz-neural-decision-signals

S7. Prespecified Control Analyses

S7.1 Stimulus-Side Decoding

On unequal-contrast trials, the target is $g_k = 1$ when right contrast exceeds left contrast and 0 for the reverse. The same feature extraction, matched dimensionality, pipeline, balanced-accuracy score, and mouse-level inference are used as in the primary choice analysis.

S7.2 Difficulty-Stratified Choice Evaluation

Out-of-fold predictions from the all-trial choice model are evaluated separately for hard (absolute contrast difference 0.25), medium (0.50), and easy (≥0.75) trials. Evaluating one common model avoids fitting a separate high-variance decoder within each smaller stratum.

S7.3 Correct and Error Trials

The all-trial model's out-of-fold choice predictions are scored separately on correct and error trials. A complementary stimulus-side decoder is fitted only to correct unequal-contrast trials and then evaluated on error unequal-contrast trials. Because the stimulus-response mapping reverses on errors, this cross-condition test probes whether a learned signal follows stimulus side or response direction; its power is limited by the smaller number of errors.

S7.4 Equal-Contrast Trials

Choice decoding is repeated using trials for which left and right contrasts are equal. In this subset, signed stimulus side is undefined and cannot explain decoding, although response preparation, movement, engagement, and other choice-correlated variables remain possible contributors.

S7.5 Pre-Movement Evaluation

Out-of-fold choice predictions are evaluated on trials whose recorded reaction time is at least 250 ms for the post-early window and at least 500 ms for the post-late window. The analyzed neural window therefore ends before recorded movement onset. This restriction does not eliminate covert motor preparation.

S8. Complete Supplementary Results

S8.1 Stimulus-side decoding

Mouse-level balanced accuracy on unequal-contrast trials.

Condition	n mice	Regional rate	Inter-area	Regional - inter	Paired p_{Holm}	Regional vs 0.5 p_{Holm}	Inter vs 0.5 p_{Holm}
Pre-early stimulus side	10	49.1% [48.0, 50.3]	49.9% [49.0, 50.7]	-0.8% [-1.6, 0.2]	0.3008	0.3359	0.9062
Pre-late stimulus side	10	48.9% [47.1, 50.7]	49.5% [48.4, 50.6]	-0.7% [-2.7, 1.6]	0.5645	0.3359	0.9062
Post-early stimulus side	10	69.0% [64.7, 73.0]	54.2% [52.8, 55.6]	14.8% [11.6, 17.9]	0.0078	0.0078	0.0078
Post-late stimulus side	10	70.5% [67.0, 74.2]	50.5% [49.8, 51.2]	20.0% [16.8, 23.5]	0.0078	0.0078	0.6211

S8.2 Equal-contrast choice decoding

Choice decoding after removing signed stimulus side.

Condition	n mice	Regional rate	Inter-area	Regional - inter	Paired p_{Holm}	Regional vs 0.5 p_{Holm}	Inter vs 0.5 p_{Holm}
Pre-early choice, equal contrast	10	55.0% [49.2, 59.9]	48.0% [45.9, 50.3]	7.0% [1.2, 12.6]	0.0547	0.1602	0.4766
Pre-late choice, equal contrast	10	53.9% [50.3, 57.6]	48.8% [46.4, 51.0]	5.2% [2.1, 8.6]	0.0410	0.1602	0.6875
Post-early choice, equal contrast	10	59.1% [54.2, 64.2]	50.2% [47.0, 53.1]	8.9% [3.3, 15.3]	0.0410	0.0234	0.9277
Post-late choice, equal contrast	10	59.8% [55.6, 64.2]	48.3% [46.4, 50.2]	11.5% [7.2, 15.8]	0.0078	0.0078	0.4766

S8.3 Correct-to-error stimulus generalization

Stimulus-side decoder trained on correct unequal-contrast trials and evaluated on error unequal-contrast trials.

Condition	n mice	Regional rate	Inter-area	Regional - inter	Paired p_{Holm}	Regional vs 0.5 p_{Holm}	Inter vs 0.5 p_{Holm}
Pre-early correct-to-error stimulus	9	53.7% [44.9, 63.8]	46.5% [41.3, 51.4]	7.2% [-0.9, 15.0]	0.4219	1.0000	0.6680
Pre-late correct-to-error stimulus	9	47.6% [39.9, 54.3]	52.6% [50.2, 55.1]	-5.0% [-13.0, 2.4]	0.5156	1.0000	0.3594
Post-early correct-to-error stimulus	9	56.6% [52.3, 60.8]	46.7% [42.3, 50.8]	9.8% [3.0, 16.9]	0.1406	0.1250	0.6680
Post-late correct-to-error stimulus	9	50.7% [44.9, 57.7]	52.3% [48.1, 55.5]	-1.7% [-7.3, 4.6]	0.6211	1.0000	0.6680

S8.4 Difficulty-stratified choice decoding

Out-of-fold choice predictions evaluated within absolute contrast-difference strata.

Condition	n mice	Regional rate	Inter-area	Regional - inter	Paired p_{Holm}	Regional vs 0.5 p_{Holm}	Inter vs 0.5 p_{Holm}
Post-early choice, hard	10	59.5% [56.2, 63.1]	53.4% [50.6, 56.1]	6.1% [2.2, 9.7]	0.0234	0.0117	0.1953
Post-early choice, medium	10	67.0% [62.3, 71.8]	53.2% [51.0, 55.2]	13.8% [9.5, 18.4]	0.0117	0.0117	0.1074
Post-early choice, easy	10	70.5% [66.3, 75.1]	54.4% [53.1, 55.7]	16.1% [12.6, 19.7]	0.0117	0.0117	0.0117
Post-late choice, hard	10	65.4% [60.9, 69.1]	51.7% [48.7, 54.1]	13.7% [10.7, 17.1]	0.0117	0.0117	0.8320
Post-late choice, medium	10	71.3% [67.6, 75.4]	49.3% [47.1, 51.3]	21.9% [17.5, 27.3]	0.0117	0.0117	1.0000
Post-late choice, easy	10	71.3% [66.9, 75.9]	50.1% [49.0, 51.4]	21.2% [17.2, 25.5]	0.0117	0.0117	1.0000

S8.5 Correctness-stratified choice decoding

Out-of-fold choice predictions evaluated separately on correct and error trials.

Condition	n mice	Regional rate	Inter-area	Regional - inter	Paired p_{Holm}	Regional vs 0.5 p_{Holm}	Inter vs 0.5 p_{Holm}
Post-early choice, correct	10	67.7% [63.8, 71.8]	54.4% [53.0, 55.7]	13.3% [10.1, 16.5]	0.0078	0.0078	0.0078
Post-early choice, error	10	53.9% [50.6, 57.4]	49.9% [47.0, 52.5]	4.0% [-0.1, 7.8]	0.1035	0.0977	1.0000
Post-late choice, correct	10	71.3% [67.5, 75.4]	50.5% [49.3, 51.8]	20.8% [16.9, 25.2]	0.0078	0.0078	1.0000
Post-late choice, error	10	54.4% [50.6, 58.2]	48.8% [46.8, 50.9]	5.6% [2.0, 9.0]	0.0391	0.0977	0.9727

S8.6 Pre-movement choice decoding

Out-of-fold predictions evaluated only when recorded movement began after the analyzed window.

Condition	n mice	Regional rate	Inter-area	Regional - inter	Paired p_{Holm}	Regional vs 0.5 p_{Holm}	Inter vs 0.5 p_{Holm}
Post-early choice, movement after 250 ms	10	60.5% [57.4, 63.6]	51.8% [49.9, 53.3]	8.6% [5.5, 11.5]	0.0078	0.0039	0.1562
Post-late choice, movement after 500 ms	10	54.4% [51.3, 57.1]	49.2% [47.7, 50.5]	5.2% [2.1, 8.1]	0.0137	0.0312	0.3828

S8.7 Temporal contrasts

Paired mouse-level temporal changes and feature-family-by-time interactions.

Contrast	Feature	n mice	Mean difference	95% CI	Raw p	Holm p
Post-early minus pre-late	local	10	14.7%	[10.9, 18.7]	0.0020	0.0117
Post-late minus pre-late	local	10	17.8%	[13.6, 22.1]	0.0020	0.0117
Post-early minus pre-late	inter	10	3.3%	[1.6, 4.9]	0.0098	0.0195
Post-late minus pre-late	inter	10	0.0%	[-1.6, 1.8]	0.9863	0.9863
Feature-family difference: post-early minus pre-late	interaction	10	11.5%	[8.3, 14.9]	0.0020	0.0117
Feature-family difference: post-late minus pre-late	interaction	10	17.8%	[13.8, 22.2]	0.0020	0.0117

S8.8 Review-motivated contrasts

Prespecified regional-rate contrasts for correctness and stimulus difficulty.

Contrast	Feature	n mice	Mean difference	95% CI	Raw p	Holm p
Post-early: correct minus error	local	10	13.8%	[8.2, 19.2]	0.0059	0.0117
Post-late: correct minus error	local	10	16.9%	[11.3, 22.5]	0.0020	0.0078
Post-early: easy minus hard	local	10	11.0%	[6.3, 15.6]	0.0039	0.0117
Post-late: easy minus hard	local	10	5.9%	[1.3, 10.3]	0.0449	0.0449

Accuracy entries are mean balanced accuracy with mouse-bootstrap 95% confidence intervals. Difference entries are percentage-point differences. Chance and paired tests are exact two-sided mouse-level sign-flip tests; adjusted values use Holm correction within the indicated family.